

No Uniform Wasserstein Convexity Modulus for Vanishing MMD Regularization

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Status

Candidate counterexample, likely valid; full negative answer in the stated generality. We give standard admissible data for which every generalized-geodesic convexity modulus tends to minus infinity as the MMD regularization parameter tends to zero.

Source Question

Stein, Neumayer, Rux, and Steidl ask in Remark 23 of *Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces*, arXiv:2402.04613 [1], whether the functionals $D_{f,\nu}^{\lambda_N}$, with $\lambda_N \downarrow 0$, have a uniform lower bound on their convexity moduli along generalized geodesics. Such a bound would permit direct application of a stability theorem for Wasserstein gradient flows.

Consequently, we obtain $\partial_t \gamma_N - \operatorname{div}(\gamma_N \nabla \hat{p}_{\gamma_N}) = 0$ as required. $\square$

Remark 23 (Consistent discretization). *An advantage of the regularized f -divergences $D_{f,\nu}^\lambda$, $\nu \in \mathcal{P}_2(\mathbb{R}^d)$, is that $D_{f,\nu}^\lambda(\mu) < \infty$ for any $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ (even if $D_{f,\nu}(\mu) = \infty$). This is important when approximating the Wasserstein gradient flow of the functional $D_{f,\nu}$ starting in $\mu_0 \in \operatorname{dom}(D_{f,\nu})$ numerically with gradient flows starting in empirical measures $\mu_{0,N} = \frac{1}{N} \sum_{i=1}^N \delta_{x_i(0)}$. Since $\mu_{0,N}$ might not be in $\operatorname{dom}(D_{f,\nu})$, we choose $(\lambda_N)_{N \in \mathbb{N}}$ with $\lim_{N \rightarrow \infty} \lambda_N = 0$ such that $\sup_{N \in \mathbb{N}} D_{f,\nu}^{\lambda_N}(\mu_{0,N}) < \infty$. Then, as shown in Corollary 13(i), the functionals $D_{f,\nu}^{\lambda_N}$ Mosco converge to $D_{f,\nu}$. If we have a uniform lower bound on their convexity-modulus along generalized geodesics, then [4, Thm. 11.2.1] implies that the Wasserstein gradient flows of $D_{f,\nu}^{\lambda_N}$ starting in $\mu_{0,N}$ converge locally uniformly in $[0, \infty)$ to the flow of $D_{f,\nu}$ starting in μ_0 . Whether such a lower bound exists is so far an open question. In Theorem 19, we only established a lower bound on the weak convexity modulus of $D_{f,\nu}^{\lambda_N}$ which scales as $1/\lambda_N$. A similar convergence result was recently established in [43] for regularization with the Wasserstein distance W_2 instead of the MMD d_K . However, their proof cannot be directly extended to our setting.*

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We use the source's convention: a functional E is κ -convex along generalized geodesics if the appropriate generalized geodesic $\gamma_{\alpha,t}$ satisfies

$$E(\gamma_{\alpha,t}) \leq (1-t)E(\mu) + tE(\eta) - \frac{\kappa}{2}t(1-t)W_{\alpha}^2(\mu, \eta).$$

Counterexample

Theorem 1. *Let $d = 1$, let*

$$K(x, y) = \exp\left(-\frac{(x - y)^2}{2}\right), \quad f(a) = (a - 1)^2 \quad (a \geq 0), \quad \nu = \delta_0,$$

where $f = +\infty$ on $(-\infty, 0)$. Put $E_\lambda = D_{f, \delta_0}^\lambda$ for $\lambda > 0$. If E_λ is κ_λ -convex along generalized geodesics, then

$$\kappa_\lambda \leq -\frac{1}{e\lambda(1 + 2\lambda)}.$$

In particular, if $\lambda_N \downarrow 0$, there is no finite uniform lower bound for the moduli κ_{λ_N} .

Proof. The function f is the standard Tsallis–2 entropy and has infinite recession constant. Therefore, for target δ_0 , a nonnegative measure σ has finite $D_f(\sigma \mid \delta_0)$ only if $\sigma = a\delta_0$ for some $a \geq 0$. For $c = K(x, 0) = e^{-x^2/2}$, the definition of the non-tight regularization gives

$$H_\lambda(x) := E_\lambda(\delta_x) = \min_{a \geq 0} \left\{ (a - 1)^2 + \frac{1 + a^2 - 2ac}{2\lambda} \right\}.$$

The unique unconstrained minimizer is

$$a_\lambda(x) = \frac{2\lambda + c}{2\lambda + 1} > 0,$$

so it also solves the constrained problem. Substitution yields

$$\begin{aligned} H_\lambda(x) &= \frac{(1 - c)(c + 4\lambda + 1)}{2\lambda(1 + 2\lambda)} \\ &= \frac{1 - e^{-x^2}}{2\lambda(1 + 2\lambda)} + \frac{2(1 - e^{-x^2/2})}{1 + 2\lambda}. \end{aligned} \tag{1}$$

The source records the same Dirac/Tsallis–2 computation in Remark 27:

Since the radial kernel K is a decreasing function of the squared norm, the optimal $a^ = K(x, y)$ is in general smaller than 1. If, additionally we require that $K(x, x) = 1$ for all $x \in \mathbb{R}$, then we have the optimal value*

$$D_{f, \delta_y}^\lambda(\delta_x) = \frac{1 - K(x, y)}{\lambda} \frac{K(x, y) + 4\lambda + 1}{2(2\lambda + 1)}.$$

For $g_a(x) = 1 - e^{-ax^2}$,

$$g_a''(x) = 2ae^{-ax^2}(1 - 2ax^2).$$

At $x = 1$ this gives $g_1''(1) = -2/e$ and $g_{1/2}''(1) = 0$. Hence (1) implies

$$H_\lambda''(1) = -\frac{1}{e\lambda(1 + 2\lambda)}. \tag{2}$$

It remains to relate this one-dimensional curvature to generalized Wasserstein geodesics. Fix $h > 0$ and take endpoints δ_{1-h} and δ_{1+h} . In every three-plan defining a generalized geodesic between these endpoints, the second and third coordinates are fixed. Consequently the generalized geodesic is necessarily

$$t \longmapsto \delta_{1+(2t-1)h}, \quad W_\alpha^2 = 4h^2,$$

independently of its base. Applying κ_λ -convexity at $t = 1/2$ gives

$$H_\lambda(1+h) + H_\lambda(1-h) - 2H_\lambda(1) \geq \kappa_\lambda h^2.$$

After division by h^2 and passage to $h \downarrow 0$, equation (2) gives

$$\kappa_\lambda \leq H''_\lambda(1) = -\frac{1}{e\lambda(1+2\lambda)}.$$

The right-hand side tends to $-\infty$ as $\lambda \downarrow 0$. □

The Source's Approximation Hypotheses Still Hold

This obstruction is compatible with the exact setup of Remark 23. Set

$$\lambda_N = \frac{1}{N}, \quad \mu_0 = \delta_0, \quad \mu_{0,N} = \frac{1}{N} \sum_{i=1}^N \delta_0 = \delta_0.$$

The initial measures are empirical, converge to μ_0 , and satisfy

$$\sup_N D_{f,\delta_0}^{\lambda_N}(\mu_{0,N}) = 0.$$

Nevertheless, Theorem 1 forces every valid global modulus to obey

$$\kappa_{\lambda_N} \leq -\frac{N}{e(1+2/N)} \longrightarrow -\infty.$$

Thus the requested uniform lower bound does not exist even when all the energy and empirical-approximation assumptions in the source remark are met.

Scope and Interpretation

The theorem answers the source's question negatively under its general assumptions. It does not assert that these gradient flows fail to converge: for the particular initial data above the flows are stationary. What fails is the proposed global uniform-semiconvexity route to convergence. Uniform bounds under additional restrictions—for example non-atomic targets, selected entropy/kernel classes, or along controlled trajectories—are not addressed here.

Verification and Novelty Bounds

The scalar minimization and the value of $H''_\lambda(1)$ were independently checked symbolically. The full algebra is recorded in `verification.md`. The cheap run indexes contain no prior entry for arXiv:2402.04613 or for this Gaussian/Tsallis-2 curvature obstruction. Bounded arXiv searches on 2026-07-21 found the source and related MMD-flow papers, but no later explicit resolution of this uniform-modulus question.

References

- [1] V. Stein, S. Neumayer, N. Rux, and G. Steidl. *Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces*. arXiv:2402.04613, 2024; revised 2025. arXiv:2402.04613.